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322652(22)

B. E. (Sixth Semester) Examination,
Nov.-Dec. 2015

(New Scheme)

(CSE Engg. Branch)

COMPILER DESIGN

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *All questions carry equal marks. Subquestion (a) is compulsory. Attempt any two subparts from (b), (c) & (d).*

1. (a) Consider the following program. 2

```
void main( )
{
int x, y;
```


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```
float s, z;
y = x + 1;
z = s + 1;
}
```

List down the leximes, tokens and the attributes of the tokens, at the end of the Lexical analysis of the above program. Also find the number of tokens.

(b) Describe the various phases of compiler.

(c) What are the different compiler construction tools? 7

(d) Write a LEX program specification for the Tokens defined by identifier / tokens are BEGIN / END ? THEN / ELSE / letter (letter/digit) * / digit + / < / < = / = / < > / > / > =. 7

2. (a) What is handle and handle pruning in case of Bottom-up parsing? 2

(b) Consider the following grammar :

$$E \rightarrow E+T/T,\ T \rightarrow TF/F,\ F \rightarrow F^*|a|b$$

construct predictive parsing table for this grammar. Is this grammer, LL(1) grammer. 7

(c) Contruct an LARL(,) parsing table for the following

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grammar :

$S \rightarrow Aa | bAC | dc | bda, A \rightarrow d$

(d) Write a Yacc 'desk calculate' program that will evaluate Boolean expressions. 7

3. (a) What is syntax directed translation scheme? 2

(b) Using the given grammar, write the syntax directed definition to evaluate an expression. Construct the parse tree for these sentence 2+3*7

$E \rightarrow E + T \mid T$

$T \rightarrow T * F \mid F$

$F \rightarrow (E) \mid \text{Num}$ 7

(c) Consider the translation scheme shown below : 7

$E \rightarrow E + T \mid T$

$T \rightarrow \text{num}$

write the SDD for the postfix evaluation. For an input 2 + 3 + 4, what this translation scheme prints.

(d) Translate the following expression to quadruple, triple and indirect triple 7

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$-(x+y)*(z+c)-(x+y+z)$

4. (a) Write the basic data structure that can be used to implement symbol tables. 2

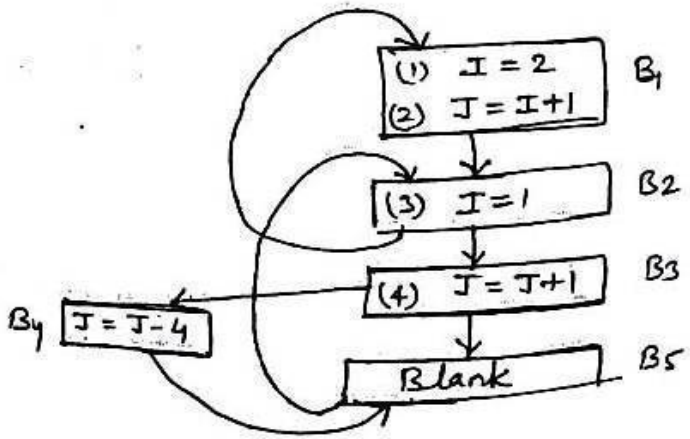
(b) Explain various allocation strategies in detail. 7

(c) What is parameter parsing techniques? Explain various methods of parameter parsing. 7

(d) What is the use of activation record? Explain different fields in the activation record. 7

5. (a) Define code optimization. 2

(b) Consider, the following flow graph, compute the data flow equations and count the number of passes. 7



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(c) Discuss various issues in design of a code generation. 7

(d) Generate assembly code for the following 'C' statement :

$$x = a|(b+c) - d*(e+f)$$ 7

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